

1.Importación y limpieza del dataset

```
In [73]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [ ]:
```

```
In [74]: ruta = "2024_Accidentalidad.csv"
df = pd.read_csv(ruta, sep=';', encoding='utf-8')
```

```
In [75]: df.head(2)
```

```
Out[75]:
```

	num_expediente	fecha	hora	localizacion	numero	cod_distrito	distrito
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0	2023S040280	04/01/2024	14:09:00	AVDA. NICETO ALCALA ZAMORA / AUTOV. M- 11	3	16	HORTALEZA
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1	2023S040280	04/01/2024	14:09:00	AVDA. NICETO ALCALA ZAMORA / AUTOV. M- 11	3	16	HORTALEZA
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```
In [76]: df1 = df.copy()
```

```
In [77]: df1.shape
```

```
Out[77]: (49340, 19)
```

```
In [78]: pd.DataFrame({
    'Tipo de dato': df1.dtypes,
    'Valores no nulos': df1.count()
})
```

Out[78]:

	Tipo de dato	Valores no nulos
num_expediente	object	49340
fecha	object	49340
hora	object	49340
localizacion	object	49340
numero	object	49340
cod_distrito	int64	49340
distrito	object	49340
tipo_accidente	object	49335
estado_meteorológico	object	43424
tipo_vehiculo	object	48946
tipo_persona	object	49340
rango_edad	object	49340
sexo	object	49340
cod_lesividad	float64	27344
lesividad	object	27344
coordenada_x_utm	float64	49334
coordenada_y_utm	float64	49334
positiva_alcohol	object	49173
positiva_droga	float64	203

```
In [79]: pd.DataFrame({  
    'Tipo de dato': df1.dtypes,  
    'Valores no nulos': df1.isna().sum()  
})
```

Out[79]:

	Tipo de dato	Valores no nulos
num_expediente	object	0
fecha	object	0
hora	object	0
localizacion	object	0
numero	object	0
cod_distrito	int64	0
distrito	object	0
tipo_accidente	object	5
estado_meteorológico	object	5916
tipo_vehiculo	object	394
tipo_persona	object	0
rango_edad	object	0
sexo	object	0
cod_lesividad	float64	21996
lesividad	object	21996
coordenada_x_utm	float64	6
coordenada_y_utm	float64	6
positiva_alcohol	object	167
positiva_droga	float64	49137

In [80]: `df1.drop(['localizacion', 'numero', 'coordenada_x_utm', 'coordenada_y_utm'], axis=1)`In [81]: `df1.head(2)`

Out[81]:

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado_meteorológico
0	2023S040280	04/01/2024	14:09:00	16	HORTALEZA	Colisión fronto-lateral	
1	2023S040280	04/01/2024	14:09:00	16	HORTALEZA	Colisión fronto-lateral	

In [82]: `df2 = df1.copy()`In [83]: `list(df2['num_expediente'].unique())`

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```

```
In [84]: df2['num_expediente'].str.startswith('2023S').sum()
```

```
Out[84]: 11
```

```
In [85]: df2 = df2.loc[~df2['num_expediente'].str.startswith('2023S')]
```

```
In [86]: df2['num_expediente'].str.startswith('2023S').sum()
```

```
Out[86]: 0
```

```
In [87]: list(df2['fecha'].unique())
```

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```

```
In [88]: df2['fecha'] = pd.to_datetime(df2['fecha'], format='%d/%m/%Y')
```

```
In [89]: df2.dtypes
```

```
Out[89]: num_expediente      object  
fecha          datetime64[ns]  
hora           object  
cod_distrito    int64  
distrito        object  
tipo_accidente  object  
estado_meteorológico object  
tipo_vehiculo   object  
tipo_persona    object  
rango_edad      object  
sexo            object  
cod_lesividad   float64  
lesividad       object  
positiva_alcohol object  
positiva_droga  float64  
dtype: object
```

```
In [90]: df3 = df2.copy()
```

```
In [91]: df3['hora'] = pd.to_datetime(df3['hora'], format='%H:%M:%S').dt.time
```

```
In [92]: list(df3['hora'].unique())
```

```
Out[92]: [datetime.time(1, 20),
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datetime.time(9, 17),
datetime.time(20, 19),
datetime.time(23, 9),
datetime.time(8, 51),
datetime.time(8, 53),
datetime.time(10, 59),
datetime.time(16, 51),
datetime.time(19, 1),
datetime.time(8, 32),
datetime.time(9, 18),
datetime.time(19, 49),
datetime.time(18, 6),
datetime.time(19, 29),
datetime.time(1, 24),
datetime.time(10, 18),
datetime.time(13, 36),
datetime.time(15, 14),
datetime.time(23, 14),
datetime.time(1, 32),
datetime.time(3, 4),
...]

```

In [93]: `from datetime import time`

```

def clasificar_horario(hora):
    if time(7, 0) <= hora < time(15, 0):
        return 'Mañana'
    elif time(15, 0) <= hora < time(23, 0):
        return 'Tarde'
    else:
        return 'Noche'

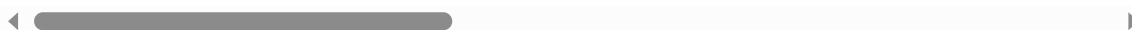
```

In [94]: `df3['franja_horaria'] = df3['hora'].apply(clasificar_horario)`

In [95]: `df3.tail()`

Out[95]:

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estad
49335	2024S035474	2024-12-23	11:35:00	20	SAN BLAS-CANILLEJAS	Otro	
49336	2024S035479	2024-12-03	09:45:00	9	MONCLOA-ARAVACA	Otro	
49337	2024S035479	2024-12-03	09:45:00	9	MONCLOA-ARAVACA	Otro	
49338	2024S035480	2024-12-29	19:15:00	3	RETIRO	Colisión fronto-lateral	
49339	2024S035480	2024-12-29	19:15:00	3	RETIRO	Colisión fronto-lateral	



In [96]: df3.dtypes

```
Out[96]: num_expediente      object
fecha          datetime64[ns]
hora           object
cod_distrito    int64
distrito        object
tipo_accidente  object
estado_meteorológico  object
tipo_vehiculo   object
tipo_persona    object
rango_edad      object
sexo            object
cod_lesividad   float64
lesividad       object
positiva_alcohol object
positiva_droga  float64
franja_horaria  object
dtype: object
```

In [97]: df4= df3.copy()

In [98]: list(df4['distrito'].unique())

```
Out[98]: ['RETIRO',
          'PUENTE DE VALLECAS',
          'MORATALAZ',
          'CARABANCHEL',
          'CENTRO',
          'SALAMANCA',
          'CHAMBERÍ',
          'MONCLOA-ARAVACA',
          'CHAMARTÍN',
          'VICÁLVARO',
          'ARGANZUELA',
          'FUENCARRAL-EL PARDO',
          'USERA',
          'VILLA DE VALLECAS',
          'TETUÁN',
          'BARAJAS',
          'LATINA',
          'VILLAVERDE',
          'SAN BLAS-CANILLEJAS',
          'HORTALEZA',
          'CIUDAD LINEAL']
```

```
In [99]: list(df4['cod_distrito'].unique())
```

```
Out[99]: [3, 13, 14, 11, 1, 4, 7, 9, 5, 19, 2, 8, 12, 18, 6, 21, 10, 17, 20, 16, 15]
```

```
In [100... distrito_codigos = {
    1: 28013,
    2: 28045,
    3: 28007,
    4: 28001,
    5: 28002,
    6: 28020,
    7: 28010,
    8: 28029,
    9: 28008,
    10: 28044,
    11: 28025,
    12: 28026,
    13: 28038,
    14: 28030,
    15: 28037,
    16: 28043,
    17: 28021,
    18: 28031,
    19: 28032,
    20: 28022,
    21: 28042
}

df4['codigo_postal'] = df4['cod_distrito'].map(distrito_codigos)
```

```
In [101... df4.head(3)
```

Out[101...

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado_mete
11	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
12	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
13	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I

In [102...

df5= df4.copy()

In [103...

df5['tipo_accidente'].value_counts(dropna=False)

Out[103...

```

tipo_accidente
Colisión fronto-lateral    13241
Alcance                    12091
Colisión lateral           8013
Choque contra obstáculo fijo  5332
Atropello a persona        3042
Caída                      2678
Colisión múltiple          2532
Colisión frontal           1094
Otro                       793
Solo salida de la vía      249
Vuelco                     182
Atropello a animal         77
NaN                         5
Name: count, dtype: int64

```

In [104...

df5[df5['tipo_accidente'].isna()]['num_expediente']

Out[104...

```

45370    2024S033213
48148    2024S034789
48149    2024S034789
48150    2024S034789
48151    2024S034789
Name: num_expediente, dtype: object

```

In [105...

df5[df5['num_expediente'].isin(['2024S033213', '2024S034789'])]

Out[105...

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado
45370	2024S033213	2024-12-05	02:45:00	13	PUENTE DE VALLECAS	NaN	
48148	2024S034789	2024-12-22	06:10:00	1	CENTRO	NaN	
48149	2024S034789	2024-12-22	06:10:00	1	CENTRO	NaN	
48150	2024S034789	2024-12-22	06:10:00	1	CENTRO	NaN	
48151	2024S034789	2024-12-22	06:10:00	1	CENTRO	NaN	

In [106...

```
df5.dropna(subset=['tipo_accidente'], inplace=True)
```

In [107...

```
df5['tipo_accidente'].value_counts(dropna=False)
```

Out[107...

```

tipo_accidente
Colisión fronto-lateral    13241
Alcance                    12091
Colisión lateral           8013
Choque contra obstáculo fijo  5332
Atropello a persona        3042
Caída                      2678
Colisión múltiple          2532
Colisión frontal           1094
Otro                       793
Solo salida de la vía       249
Vuelco                     182
Atropello a animal         77
Name: count, dtype: int64

```

In [108...

```

grupo_accidente = {
    'Colisión fronto-lateral': 'Colisión',
    'Colisión lateral': 'Colisión',
    'Colisión múltiple': 'Colisión',
    'Colisión frontal': 'Colisión',
    'Alcance': 'Colisión',

    'Choque contra obstáculo fijo': 'Choque/Obstáculo/Salida via',
    'Solo salida de la vía': 'Choque/Obstáculo/Salida via',
    'Vuelco': 'Choque/Obstáculo/Salida via',

    'Atropello a persona': 'Atropello',
    'Atropello a animal': 'Atropello',

    'Caída': 'Caída',

    'Otro': 'Otro'
}

```

```
In [109... df5['clasificacion_accidente'] = df5['tipo_accidente'].map(grupo_accidente)
```

```
In [110... df5['clasificacion_accidente'] = df5['clasificacion_accidente'].fillna('Otro')
```

```
In [111... df5.head(2)
```

```
Out[111...      num_expediente  fecha      hora  cod_distrito  distrito  tipo_accidente  estado_mete
```

11	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
-----------	-------------	------------	----------	---	--------	-------------------	---

12	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
-----------	-------------	------------	----------	---	--------	-------------------	---

◀  ▶

```
In [112... df6= df5.copy()
```

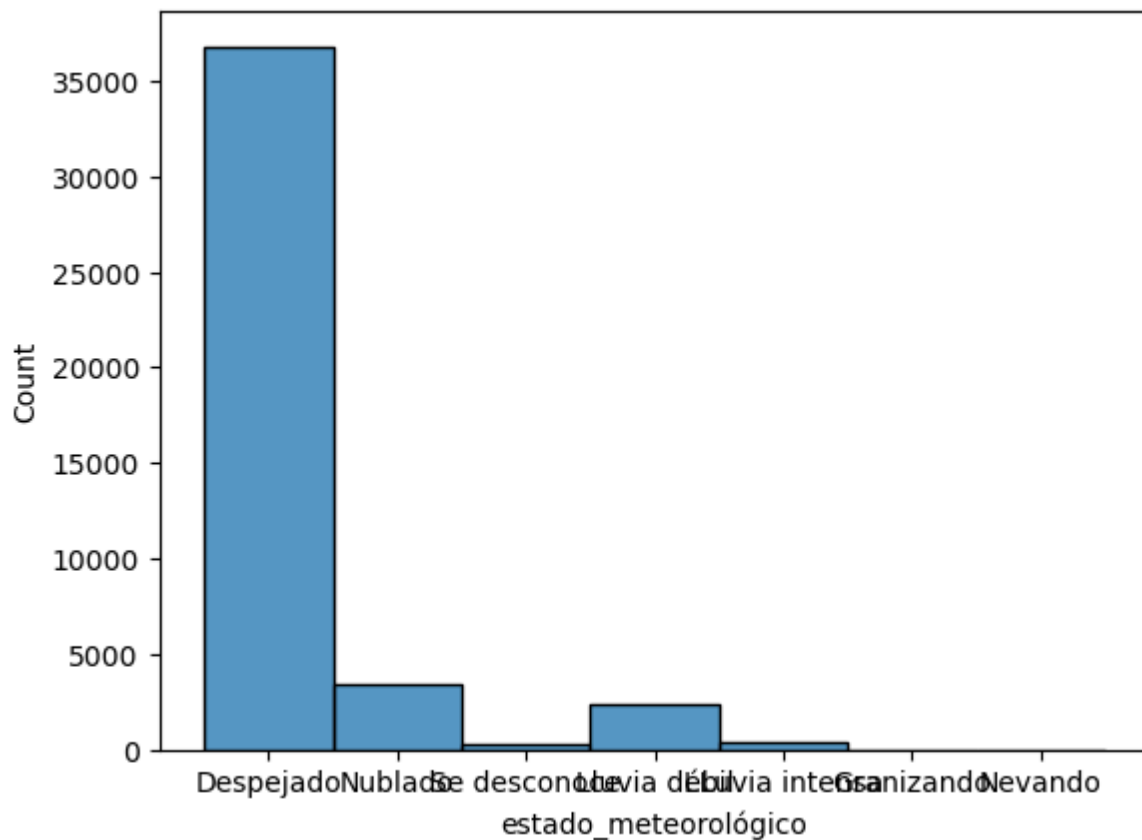
```
In [113... df6['estado_meteorológico'] = df5['estado_meteorológico']
```

```
In [114... df6['estado_meteorológico'].value_counts(dropna=False)
```

```
Out[114... estado_meteorológico
Despejado          36731
NaN                 5910
Nublado            3464
Lluvia débil       2389
LLuvia intensa     464
Se desconoce       356
Granizando         8
Nevando            2
Name: count, dtype: int64
```

```
In [115... sns.histplot(data= df6, x= 'estado_meteorológico')
```

```
Out[115... <Axes: xlabel='estado_meteorológico', ylabel='Count'>
```



```
In [116...] moda = df6['estado_meteorológico'].mode()[0]
df6['estado_meteorológico'] = df6['estado_meteorológico'].fillna(modas)
```

```
In [117...] df6['estado_meteorológico'].value_counts(dropna=False)
```

```
Out[117...] estado_meteorológico
Despejado      42641
Nublado         3464
Lluvia débil   2389
Lluvia intensa    464
Se desconoce     356
Granizando        8
Nevando          2
Name: count, dtype: int64
```

```
In [118...] df7= df6.copy()
```

```
In [119...] df7['tipo_vehículo'].value_counts(dropna=False)
```

```
Out[119...] tipo_vehiculo
Turismo 33865
Motocicleta hasta 125cc 3307
Furgoneta 3235
Motocicleta > 125cc 1804
Autobús 1501
Camión rígido 1163
Todo terreno 829
Bicicleta 811
VMU eléctrico 771
Ciclomotor 471
NaN 394
Bicicleta EPAC (pedaleo asistido) 264
Tractocamión 212
Maquinaria de obras 181
Otros vehículos con motor 164
Vehículo articulado 84
Autobús articulado 78
Cuadriciclo ligero 42
Sin especificar 37
Autobus EMT 27
Cuadriciclo no ligero 19
Ciclo 17
Autocaravana 14
Otros vehículos sin motor 12
Patinete no eléctrico 8
Moto de tres ruedas > 125cc 3
Moto de tres ruedas hasta 125cc 2
Remolque 2
Ambulancia SAMUR 2
Ciclomotor de tres ruedas 2
Semiremolque 1
Ciclo de motor L1e-A 1
Ciclomotor de dos ruedas L1e-B 1
Name: count, dtype: int64
```

```
In [120...] df7['tipo_vehiculo'] = df7['tipo_vehiculo'].fillna('Sin especificar')
```

```
In [121...] df7['tipo_vehiculo'].value_counts(dropna=False)
```

```
Out[121...] tipo_vehiculo
Turismo 33865
Motocicleta hasta 125cc 3307
Furgoneta 3235
Motocicleta > 125cc 1804
Autobús 1501
Camión rígido 1163
Todo terreno 829
Bicicleta 811
VMU eléctrico 771
Ciclomotor 471
Sin especificar 431
Bicicleta EPAC (pedaleo asistido) 264
Tractocamión 212
Maquinaria de obras 181
Otros vehículos con motor 164
Vehículo articulado 84
Autobús articulado 78
Cuadriciclo ligero 42
Autobus EMT 27
Cuadriciclo no ligero 19
Ciclo 17
Autocaravana 14
Otros vehículos sin motor 12
Patinete no eléctrico 8
Moto de tres ruedas > 125cc 3
Moto de tres ruedas hasta 125cc 2
Remolque 2
Ambulancia SAMUR 2
Ciclomotor de tres ruedas 2
Semiremolque 1
Ciclo de motor L1e-A 1
Ciclomotor de dos ruedas L1e-B 1
Name: count, dtype: int64
```

```
In [122...] grupo_vehiculo = {

    'Bicicleta': 'Sin motor',
    'Ciclo': 'Sin motor',
    'Patinete no eléctrico': 'Sin motor',

    'Bicicleta EPAC (pedaleo asistido)': 'Ligero eléctrico',
    'VMU eléctrico': 'Ligero eléctrico',

    'Ciclomotor': 'Moto/Ciclomotor',
    'Ciclomotor de dos ruedas L1e-B': 'Moto/Ciclomotor',
    'Ciclomotor de tres ruedas': 'Moto/Ciclomotor',
    'Ciclo de motor L1e-A': 'Moto/Ciclomotor',
    'Motocicleta hasta 125cc': 'Moto/Ciclomotor',
    'Motocicleta > 125cc': 'Moto/Ciclomotor',
    'Moto de tres ruedas hasta 125cc.': 'Moto/Ciclomotor',
    'Moto de tres ruedas > 125cc': 'Moto/Ciclomotor',

    'Turismo': 'Turismo/Furgoneta',
    'Furgoneta': 'Turismo/Furgoneta',
    'Autocaravana': 'Turismo/Furgoneta',
    'Todo terreno': 'Turismo/Furgoneta',
```

```

'Autobús': 'Transporte colectivo',
'Autobús articulado': 'Transporte colectivo',
'Autobus EMT': 'Transporte colectivo',
'Ambulancia SAMUR': 'Transporte colectivo',

'Camión rígido': 'Pesado/Especial',
'Cuadriciclo ligero': 'Pesado/Especial',
'Cuadriciclo no ligero': 'Pesado/Especial',
'Vehículo articulado': 'Pesado/Especial',
'Tractocamión': 'Pesado/Especial',
'Remolque': 'Pesado/Especial',
'Semirremolque': 'Pesado/Especial',
'Maquinaría de obras': 'Pesado/Especial',

'Otros vehículos con motor': 'Otro',
'Otros vehículos sin motor': 'Otro',
'Sin especificar': 'Otro'
}

```

```
In [123... df7['clasificacion_vehiculo'] = df5['tipo_vehiculo'].map(grupo_vehiculo).fillna(
```

```
In [124... df7['clasificacion_vehiculo'].value_counts()
```

```
Out[124... clasificacion_vehiculo
Turismo/Furgoneta      37943
Moto/Ciclomotor        5589
Pesado/Especial        1703
Transporte colectivo   1608
Ligero eléctrico      1035
Sin motor              836
Otro                   610
Name: count, dtype: int64
```

```
In [125... df7.head()
```

Out[125...

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado_mete
11	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
12	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
13	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
14	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
15	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I



In [126...

```
df8= df7.copy()
```

In [127...

```
df8['tipo_persona'].value_counts(dropna=False)
```

Out[127...

```

tipo_persona
Conductor      39549
Pasajero       8153
Peatón         1622
Name: count, dtype: int64

```

In [128...

```
df8['rango_edad'].value_counts(dropna=False)
```

Out[128...

```

rango_edad
Desconocido      5305
De 45 a 49 años  5002
De 30 a 34 años  4828
De 25 a 29 años  4811
De 40 a 44 años  4697
De 35 a 39 años  4584
De 50 a 54 años  4538
De 55 a 59 años  3485
De 21 a 24 años  3276
De 60 a 64 años  2436
De 18 a 20 años  1577
Más de 74 años   1207
De 65 a 69 años  1172
De 70 a 74 años   728
De 15 a 17 años   466
Menor de 5 años   458
De 10 a 14 años   443
De 6 a 9 años     311
Name: count, dtype: int64

```

```
In [129... def clasificar_edad(valor):
    if pd.isna(valor) or valor == 'Desconocido':
        return 'Desconocido'
    elif 'Menor de 5 años' in valor or 'De 6 a 9 años' in valor or 'De 10 a 14':
        return 'Menores de edad'
    elif 'De 18 a 20 años' in valor or 'De 21 a 24 años' in valor or 'De 25 a 29 años':
        return 'Hasta 29 años'
    elif 'De 30 a 34 años' in valor or 'De 35 a 39 años' in valor or 'De 40 a 44 años':
        return 'De 30 a 59 años'
    elif 'De 45 a 49 años' in valor or 'De 50 a 54 años' in valor or 'De 55 a 59 años':
        return 'De 30 a 59 años'
    elif 'De 60 a 64 años' in valor or 'De 65 a 69 años' in valor or 'De 70 a 74 años':
        return 'Mayores de 60 años'
    else:
        return 'Desconocido'

df8['grupos_por_edades'] = df8['rango_edad'].apply(clasificar_edad)
```

```
In [130... df8[['rango_edad', 'grupos_por_edades']]]
```

```
Out[130...      rango_edad  grupos_por_edades
11  De 25 a 29 años      Hasta 29 años
12  De 25 a 29 años      Hasta 29 años
13  De 30 a 34 años      De 30 a 59 años
14  De 55 a 59 años      De 30 a 59 años
15  De 18 a 20 años      Hasta 29 años
...      ...      ...
49335 De 25 a 29 años      Hasta 29 años
49336 De 40 a 44 años      De 30 a 59 años
49337 De 50 a 54 años      De 30 a 59 años
49338 De 35 a 39 años      De 30 a 59 años
49339 De 45 a 49 años      De 30 a 59 años
```

49324 rows × 2 columns

```
In [131... df8['grupos_por_edades'].value_counts(dropna=False)
```

```
Out[131... grupos_por_edades
De 30 a 59 años      27134
Hasta 29 años        9664
Mayores de 60 años   5543
Desconocido          5305
Menores de edad      1678
Name: count, dtype: int64
```

```
In [132... df8.head(2)
```

Out[132...

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado_mete
--	----------------	-------	------	--------------	----------	----------------	-------------

11	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
----	-------------	------------	----------	---	--------	-------------------	---

12	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
----	-------------	------------	----------	---	--------	-------------------	---



In [133... df8['sexo'].value_counts(dropna=False)

Out[133...
 sexo
 Hombre 29799
 Mujer 14294
 Desconocido 5231
 Name: count, dtype: int64

In [134... df8 = df8.drop(columns=['cod_lesividad'])

In [135... df9 = df8.copy()

In [136... df9['lesividad'].value_counts(dropna=False)

Out[136...
 lesividad
 NaN 21988
 Sin asistencia sanitaria 16111
 Asistencia sanitaria sólo en el lugar del accidente 5788
 Ingreso inferior o igual a 24 horas 1864
 Atención en urgencias sin posterior ingreso 1337
 Asistencia sanitaria inmediata en centro de salud o mutua 1116
 Ingreso superior a 24 horas 567
 Asistencia sanitaria ambulatoria con posterioridad 522
 Fallecido 24 horas 29
 Se desconoce 2
 Name: count, dtype: int64

In [137... df9['lesividad'] = df9['lesividad'].fillna('Se desconoce')

In [138...
 def clasificar_lesividad(valor):
 if valor == 'Se desconoce':
 return 'Se desconoce'
 elif valor == 'Sin asistencia sanitaria':
 return 'Sin asistencia sanitaria'
 elif valor == 'Fallecido 24 horas':
 return 'Fallecido 24 horas'
 else:
 return 'Con asistencia sanitaria'

In [139... df9['clasificacion_lesividad'] = df9['lesividad'].apply(clasificar_lesividad)

In [140... df9.head(2)

Out[140...

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado_mete
11	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
12	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I



In [141...

```
df9['positiva_alcohol'].value_counts(dropna=False)
```

Out[141...

```
positiva_alcohol
N      47724
S       1433
NaN      167
Name: count, dtype: int64
```

In [142...

```
df9['positiva_alcohol'] = df9['positiva_alcohol'].fillna('N')
```

In [143...

```
df9['positiva_alcohol'].value_counts(dropna=False)
```

Out[143...

```
positiva_alcohol
N      47891
S       1433
Name: count, dtype: int64
```

In [144...

```
df9['positiva_droga'].value_counts(dropna=False)
```

Out[144...

```
positiva_droga
NaN      49122
1.0       202
Name: count, dtype: int64
```

In [145...

```
df9['positiva_droga'] = df9['positiva_droga'].fillna('N')
```

In [146...

```
df9['positiva_droga'] = df9['positiva_droga'].replace(1.0, 'S')
```

In [147...

```
df9['positiva_droga'].value_counts(dropna=False)
```

Out[147...

```
positiva_droga
N      49122
S        202
Name: count, dtype: int64
```

In [148...

```
df9.head(2)
```

Out[148...

	num_expediente	fecha	hora	cod_distrito	distrito	tipo_accidente	estado_mete
11	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I
12	2024S000001	2024-01-01	01:20:00	3	RETIRO	Colisión múltiple	I



In [149...

```
df9.isnull().sum()
```

Out[149...

```
num_expediente      0
fecha                0
hora                 0
cod_distrito         0
distrito             0
tipo_accidente       0
estado_meteorológico 0
tipo_vehiculo        0
tipo_persona         0
rango_edad           0
sexo                 0
lesividad            0
positiva_alcohol     0
positiva_droga       0
franja_horaria       0
codigo_postal        0
clasificacion_accidente 0
clasificacion_vehiculo 0
grupos_por_edades    0
clasificacion_lesividad 0
dtype: int64
```

In [150...

```
df9.to_csv('df9_pfb1.csv', index=False, encoding='utf-8')
```